

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Groups as directed.*

Group – A

(Objective Type Questions)

(Compulsory)

1. Select the most suitable answer from these given or Fill in the blanks : $1 \times 10 = 10$

(a) Burning solid waste is not recommended because :

- (i) It's very expensive
- (ii) It requires a lot of space

- (iii) This requires modern technologies
- (iv) It causes many environmental issues
- (b) In which method of disposal of municipal solid waste, waste is dumped in the soil ?
 - (i) Incineration (ii) Land Filing
 - (iii) Composting (iv) Shredding
- (c) Which gas is generated on the decomposition of the organic matter present in the sanitary landfill ?
 - (i) Oxygen
 - (ii) Methane
 - (iii) Nitrogen
 - (iv) Hydrogen
- (d) In Solid waste management, waste utilization is achieved by :
 - (i) Recover, Reclamation and Reproduce
 - (ii) Reuse, Reclamation and Recycling
 - (iii) Recover, Recycling and Reproduce
 - (iv) Reuse, Reproduce and Recycling
- (e) Which of the following wastes is called the Municipal Solid Waste (MSW) ?
 - (i) Food Wastes

- (ii) Wood Pieces
- (iii) Plastic Cans
- (iv) All of these
- (f) _____ are the large open areas used to dispose solid waste.
- (g) STP stands for _____.
- (h) Solid waste generated in our home is called domestic waste. It is commonly called as _____.
- (i) ETP stands for _____.
- (j) Biomedical waste is _____ in nature.

Group – B

(Short-answer Type Questions)

Answer any **four** questions of the following :

5×4 = 20

2. Characterization of Municipal Solid Waste (MSW)
3. Why are landfills environmentally harmful ?
4. Impact of solid waste on environment.
5. Refuse Derived Fuel (RDF).
6. Effect of industrial waste on air, water and soil.

ES – 8/3

(3)

(Turn over)

7. Different techniques used in collection, storage, transportation and disposal of solid waste.

Group – C

(Long-answer Type Questions)

Answer any **four** questions of the following :

10×4 = 40

8. What is Biomedical Waste ? Write a short account of Biomedical Waste Management.
9. Discuss about sources and generation of solid waste.
10. Discuss Solid Waste Management. Who is responsible for solid waste management ?
11. Discuss the effect of industrial and solid waste effluent discharge on water quality and aquatic life.
12. Discuss about the classification and chemical composition of solid waste.
13. What are the characterization of Municipal solid waste hazardous waste and biomedical waste ?



ES – 8/3 (500)

(4) Voc(Sem-IV) — BCA
(SEC – 2) SWM